

EXPERIMENT -1 : Simulation of FDMA in MATLAB

CODE IMPLEMENTATION:

```
https://matlab.mathworks.com

HOME PLOTS APPS FIGURE FORMAT
Save As Copy Figure Print Show Code Title Subtitle X-Label Y-Label Z-Label Legend
FILE MATLAB Drive
FDMA_Simulation.m x gsm_tdma_simulation.m x
1 c1;
2 clear;
3 close all;
4
5 total_bandwidth = 1000;
6 num_users = 5;
7
8 channel_bw = total_bandwidth / num_users;
9
10 snr_db = [10 15 20 25 30];
11
12 fprintf('===== FDMA SIMULATION =====\n\n');
13 fprintf('Total Bandwidth = %.0f kHz\n', total_bandwidth);
14 fprintf('Number of Users = %d\n\n', num_users);
15
16 fprintf('User\tChannel BW (kHz)\tData Rate (kbps)\tSNR (dB)\n');
17 fprintf('-----\n');
18
19 for i = 1:num_users
20     data_rate = channel_bw;
21     fprintf('%d\t%.2f\t%.2f\t%.2f\n', ...
22         i, channel_bw, data_rate, snr_db(i));
23 end
24
25 figure;
26 bar(1:num_users, channel_bw * ones(1, num_users));
27
Command Window
```

OUTPUT:

